

PART 2

Analytical Line-Fitting Soln.

$$\hat{y} = mx + b$$

$$e_i = y_i - \hat{y}_i$$

$$= y_i - (mx_i + b)$$

$$\text{Points: } (0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)$$

$$MSE(m, b) = \frac{1}{4} \sum_{i=1}^4 (y_i - (mx_i + b))^2$$

$$= \frac{1}{4} [(0.5 - b)^2 + (3.5 - 2m - b)^2 + (1.5 - m - b)^2 + (7.5 - 3m - b)^2]$$

$$\frac{\partial MSE}{\partial m} = \frac{1}{4} [-4(3.5 - 2m - b) + (-2)(1.5 - m - b) + (-6)(7.5 - 3m - b)]$$

$$= \frac{1}{4} [(-14 + 8m + 4b) + (-3 + 2m + 2b) + (-45 + 18m + 6b)]$$

$$= \frac{1}{4} (28m + 12b - 62)$$

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5 = 0 \text{ @ minimum}$$

$$\frac{\partial MSE}{\partial b} = \frac{1}{4} [(-2)(0.5 - b) + (-2)(3.5 - 2m - b) + (-2)(1.5 - m - b) + (-2)(7.5 - 3m - b)]$$

$$= \frac{1}{4} [(-1 + 2b) + (-7 + 4m + 2b) + (-3 + 2m + 2b) + (-15 + 6m + 2b)]$$

$$= \frac{1}{4} (12m + 8b - 26)$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5 = 0 \text{ @ minimum}$$

$$7m + 3b - 15.5 = 0$$

$$3m + 2b - 6.5 = 0$$

$$5m = 11.5$$

$$m = 2.3$$

$$b = -0.2$$

$$\Rightarrow \hat{y} = 2.3x - 0.2$$

x	Actual y	Predicted $\hat{y}$	Error $y - \hat{y}$	Error ²
0	0.5	-0.2	0.7	0.49
2	3.5	4.4	-0.9	0.81
1	1.5	2.1	-0.6	0.36
3	7.5	6.7	0.8	0.64

$$MSE = \sum_{i=1}^4 \frac{\text{Error}_i^2}{4}$$

$$MSE = 0.575$$

Newton-Raphson Soln.

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5$$

$$\frac{\partial^2 MSE}{\partial m^2} = 7$$

$$\frac{\partial^2 MSE}{\partial b^2} = 2$$

$$\frac{\partial^2 MSE}{\partial m \partial b} = 3$$

$$\frac{\partial^2 MSE}{\partial b \partial m} = 3$$

$$\nabla MSE = \begin{bmatrix} 7m + 3b - 15.5 \\ 3m + 2b - 6.5 \end{bmatrix}$$

$$H = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\begin{bmatrix} m_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} m_n \\ b_n \end{bmatrix} - H^{-1} \nabla MSE$$

$$\Rightarrow m_{n+1} = m_n - \frac{\partial MSE / \partial m}{\partial^2 MSE / \partial m^2}$$

$$\Rightarrow b_{n+1} = b_n - \frac{\partial MSE / \partial b}{\partial^2 MSE / \partial b^2}$$

$$\Rightarrow m_{n+1} = m_n - \frac{7m_n + 3b_n - 15.5}{7}$$

$$\Rightarrow b_{n+1} = b_n - \frac{3m_{n+1} + 2b_n - 6.5}{2}$$

assume $m_0=0, b_0=0$

$$m_1 = 0 - \frac{7(0) + 3(0) - 15.5}{7}$$

$$m_1 = 2.214$$

$$b_1 = 0 - \frac{3(2.2) + 2(0) - 6.5}{2}$$

$$b_1 = -0.0714$$

$$(m_1, b_1) \approx (2.2, -0.07)$$

$$m_2 = 2.2 - \frac{7(2.2) + 3(-0.07) - 15.5}{7}$$

$$m_2 = 2.2449$$

$$b_2 = -0.0714 - \frac{3(2.24) + 2(-0.0714) - 6.5}{2}$$

$$b_2 = -0.1173$$

$$(m_2, b_2) \approx (2.2449, -0.1173)$$

$$m_3 = 2.2449 - \frac{7(2.2449) + 3(-0.1173) - 15.5}{7}$$

$$m_3 = 2.2646$$

$$b_3 = -0.1173 - \frac{3(2.26) + 2(-0.1173) - 6.5}{2}$$

$$b_3 = -0.1469$$

$$(m_3, b_3) \approx (2.2646, -0.1469) \rightarrow \hat{y} = 2.264x - 0.1469$$

$$MSE = \frac{1}{4} [(0.5 - b)^2 + (3.5 - 2m - b)^2 + (1.5 - m - b)^2 + (7.5 - 3m - b)^2]$$

$$MSE = 0.5766$$

Parabola-Fitting Soln.

$$\hat{y} = ax^2 + bx + c$$

$$MSE(a, b, c) = \frac{1}{4} [(0.5 - c)^2 + (3.5 - 4a - 2bc)^2 + (1.5 - a - b - c)^2 + (7.5 - 9a - 3b - c)^2]$$

$$MSE(a, b, c) = 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75$$

$$\frac{\partial MSE}{\partial a} = 49a + 18b + 7c - 41.5$$

$$\frac{\partial}{\partial a}$$

$$MSE_{aa} = 49$$

$$\frac{\partial MSE}{\partial b} = 18a + 7b + 3c - 15.5$$

$$\frac{\partial}{\partial b}$$

$$MSE_{bb} = 7$$

$$\frac{\partial MSE}{\partial c} = 7a + 3b + 2c - 6.5$$

$$\frac{\partial}{\partial c}$$

$$MSE_{cc} = 2$$

$$MSE_a = 49a + 18b + 7c - 41.5 = 0 \quad ①$$

$$MSE_b = 18a + 7b + 3c - 15.5 = 0 \quad ②$$

$$MSE_c = 7a + 3b + 2c - 6.5 = 0 \quad ③$$

$$2① - 7③$$

$$49a + 15b - 37.5 = 0 \quad ④$$

$$2② - 3③$$

$$15a + 5b - 11.5 = 0 \quad ⑤$$

$$④ - 3⑤$$

$$4a - 3 = 0$$

$$a = 0.75$$

$$b = 0.05$$

$$c = 0.55$$

$$\therefore \hat{y} = 0.75x^2 + 0.05x + 0.55$$

$$H = \begin{bmatrix} 49 & 18 & 7 \\ 18 & 7 & 3 \\ 7 & 3 & 2 \end{bmatrix}$$

x	Actual y	Predicted $\hat{y}$	Error $y - \hat{y}$	Error ²
0	0.5	0.55	-0.05	0.0025
2	3.5	3.65	-0.15	0.0225
1	1.5	1.35	0.15	0.0225
3	7.5	7.45	0.05	0.0025

$$MSE = 0.0125$$

$$a_{n+1} = a_n - \frac{49a_n + 18b_n + 7c_n - 41.5}{49} \Rightarrow a_{n+1} = \frac{41.5 - 18b_n - 7c_n}{49}$$

$$b_{n+1} = b_n - \frac{18a_{n+1} + 7b_n + 3c_n - 15.5}{7} \Rightarrow b_{n+1} = \frac{15.5 - 18a_{n+1} - 3c_n}{7}$$

$$c_{n+1} = c_n - \frac{7a_{n+1} + 3b_{n+1} + 2c_n - 6.5}{2} \Rightarrow c_{n+1} = \frac{6.5 - 7a_{n+1} - 3b_{n+1}}{2}$$

$$(a_0, b_0, c_0) = (0, 0, 0)$$

$$a_1 = \frac{41.5}{49} = 0.8469$$

$$a_2 = \frac{41.5 - 18b_1 - 7c_1}{49} = 0.8005$$

$$b_1 = \frac{15.5 - 18(0.8469)}{7} = 0.0364$$

$$b_2 = \frac{15.5 - 18a_2 - 3c_1}{7} = 0.0567$$

$$c_1 = \frac{6.5 - 7(0.8469) - 3(0.0364)}{2} = 0.231$$

$$c_2 = \frac{6.5 - 7a_2 - 3b_2}{2} = 0.363$$

$$a_3 = \frac{41.5 - 18b_2 - 7c_2}{49} = 0.774$$

$$b_3 = \frac{15.5 - 18a_3 - 3c_2}{7} = 0.0678$$

$$c_3 = \frac{6.5 - 7a_3 - 3b_3}{2} = 0.4384$$

$$(a_3, b_3, c_3) \approx (0.774, 0.0678, 0.4384) \Rightarrow \hat{y} = 0.774x^2 + 0.0678x + 0.4384$$

$$MSE = \frac{1}{4} [(0.5 - c_3)^2 + (3.5 - 4a_3 - 2b_3 - c_3)^2 + (1.5 - a_3 - b_3 - c_3)^2 + (7.5 - 9a_3 - 3b_3 - c_3)^2]$$

$$MSE = 0.02333$$

Parabola Soln

	a	b	c	MSE
Analytical	0.75	0.05	0.55	0.0125
3 Newton Iterations	0.774	0.0678	0.438	0.0233